

Mechanisms of solidification

- Generally, multiple crystals begin to grow in the liquid and a polycrystalline (more than one crystal) solid forms.
- Control of solidification to create single crystals for specific application:
 - Advanced casting techniques for high temperature applications (turbines blades).

Turbinebladen

R. Reed, The superalloys: fundamentals and applications, Chapter 3: Single-crystal superalloys for blades applications, Cambridge University Press, 2006

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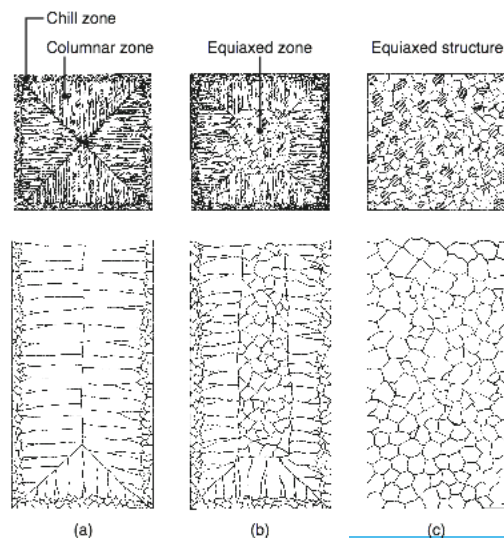
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Texture in castings

Schematic illustration of three cast structures of metals solidified in a square mold: (a) pure metals, with preferred texture at the cool mold wall. Note in the middle of the figure that only favorable oriented grains grow away from the mold surface; (b) solid-solution alloys; and (c) structure obtained by heterogeneous nucleation of grains (by adding nucleation agents).



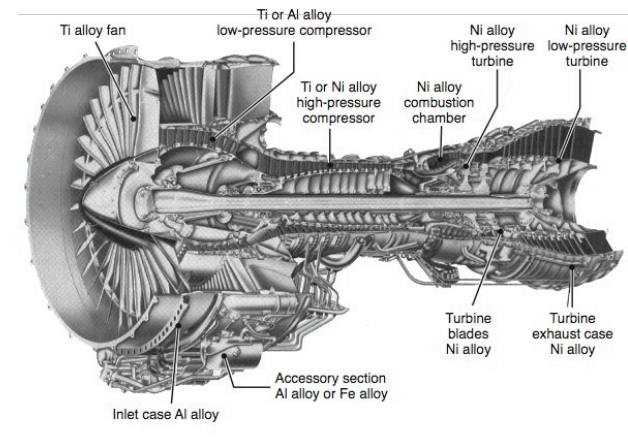
Source: Kalpakjian

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Zie hfst. 2.2.3 (Kals)

Non-ferrous alloys in aircraft engine



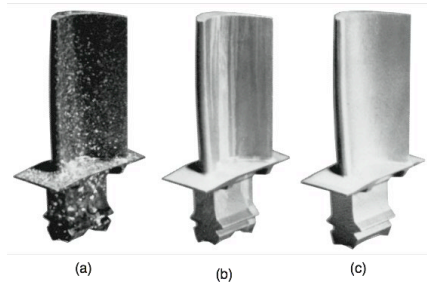
Cross-section of a jet engine (PW2037) showing various components and the alloys used in making them.

Source: Kalpakjian

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Turbine blades



Turbine blades for jet engines, manufactured by three different methods: (a) conventionally cast; (b) directionally solidified, with columnar grains, as can be seen from the vertical streaks; and (c) single crystal. Although more expensive, single-crystal blades have properties at high temperatures that are superior to those of other blades.

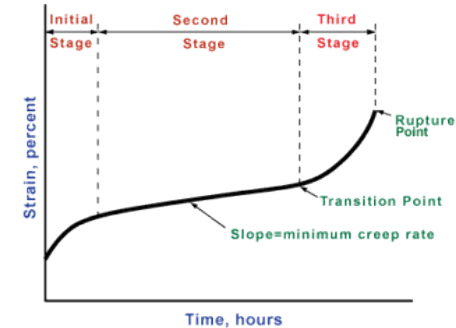
Source: Kalpakjian.

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Advantages of single crystals

- Better corrosion and creep resistance due to the absence of grain boundaries.
- What is creep?
 - High temperature progressive deformation of a material at constant stress is called creep.

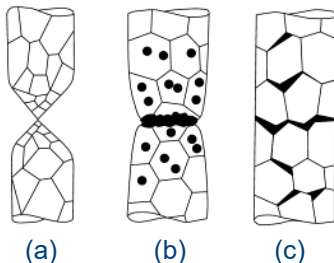


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Advantages of single crystal

- High temperature failure mechanisms in metals:



- (a) Rupture
- (b) Trans-granular fracture
- (c) Inter-granular fracture

- As temperature increase, failure mechanisms in metal undergo a transition from trans-granular to inter-granular.
- Less or no grain boundaries will limit inter-granular fracture.

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Conventional casting of turbine blades

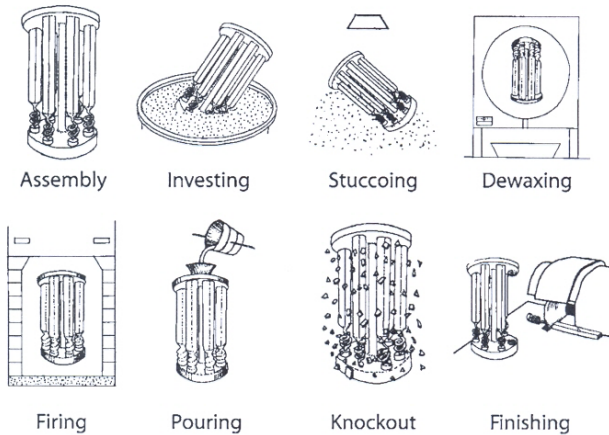
- Investment casting using a ceramic mold.
- Molten metal poured into the mold and begins solidifying against the ceramic walls.
- Presence of grain boundaries (polycrystalline structure) → susceptible to creep and cracking along those grain boundaries under centrifugal forces at high temperature.

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Illustration of various stages of investment casting for hollow blades

- Investment casting = lost wax process (wax model)



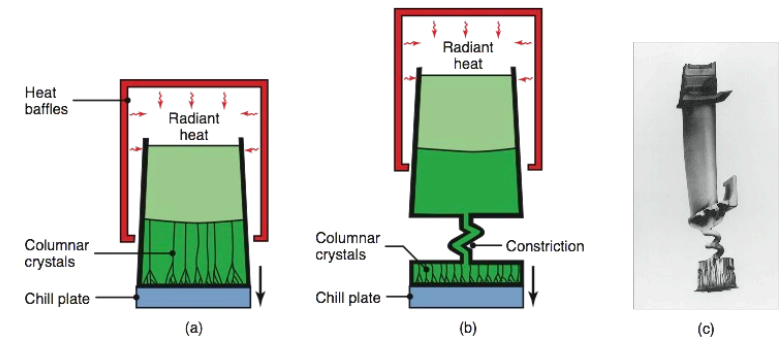
Ceramic slurries:
- Zircon ($ZrSiO_4$)
- Alumina (Al_2O_3)
- Silica (SiO_2)
- Binders

R. Reed, The superalloys: fundamentals and applications, Chapter 3: Single-crystal superalloys for blades applications, Cambridge University Press, 2006

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Turbine blade casting



Methods of casting turbine blades: (a) directional solidification; (b) method to produce a single-crystal blade; and (c) a single-crystal blade with the constriction portion still attached.

Source: Kalpakjian

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Directionally solidified blades

- Ceramic mold prepared by investment casting techniques is pre-heated by radiant heating.
- Molten superalloy is poured under vacuum at a temperature of $\sim 1550^\circ C$.
- Mold supported by a water-cooled chilled plate. Crystal begins to grow at the chill plate.
- Presence of longitudinal grain boundaries (but not transverse) $\rightarrow$ blade stronger in the direction of the centrifugal forces developed in the gas turbine (i.e. the longitudinal direction).

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How to produce a single crystal turbine blade?

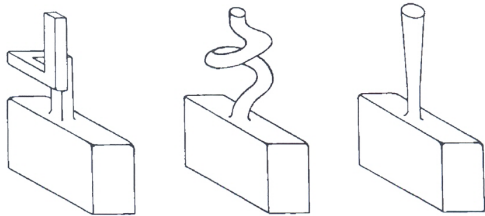
- Ceramic mold prepared by investment casting techniques but has a constriction in the shape of a spiral (grain selector). Alternatively a seed is used at the bottom of the casting.
- The assembly is lowered slowly, strict control of the rate of movement is necessary for this process.
- The cross section allow only one grain to go through $\rightarrow$ solidified mass in the mold is a single crystal.
- Absence of grain boundaries makes these blades resistant to creep and thermal shock $\rightarrow$ longer and more reliable service life.

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Single crystal blades details

- Various design of grain selector



Spiral design (centre) more efficient.

Opening not significantly larger than the typical grain size, allowing only one grain to enter the cavity.

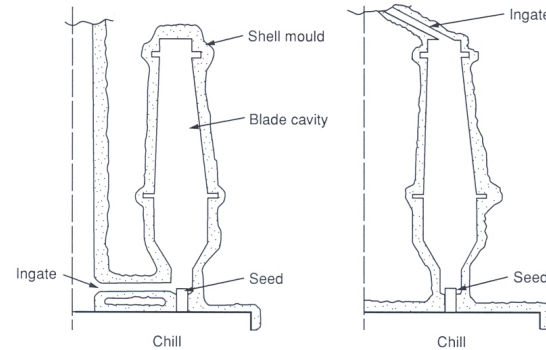
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Single crystal blades details

- Growth of single-crystal by the seeding technique



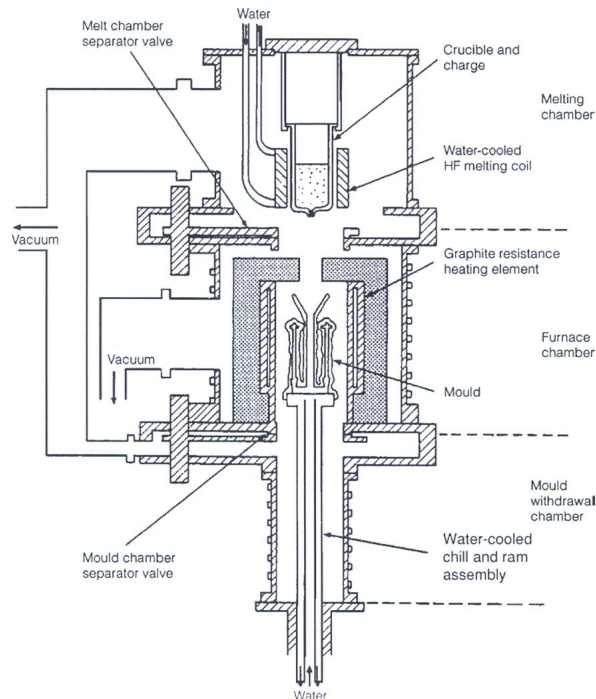
If processing conditions such that the seed is not remelted, growth occurs in a direction consistent with that of the seed.

R. Reed, The superalloys: fundamentals and applications, Chapter 3: Single-crystal superalloys for blades applications, Cambridge University Press, 2006

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Investment casting furnace for single-crystal turbine blades



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Investment casting furnace for single-crystal turbine blades

Example of furnace used by Rolls-Royce (see previous slide)

- Investment shell sits on a chill plate of about 14 cm diameter.
- 5 kg of melt required (2 minutes by HF induction melting).
- Upper chamber – central mold chamber – lower withdrawal chamber.
- Under vacuum, mold raised into the central chamber (maintained at temperature above liquidus).
- Immediately after pouring, mold is withdrawn at controlled rate. Thermal gradient typically 4000K/m.
- Cooling time between critical temperatures of 1400°C and 1000°C is typically 20 minutes.

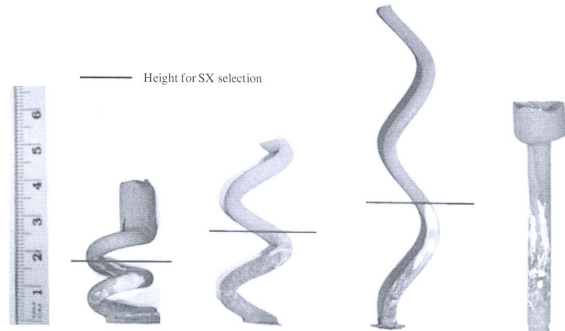
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Grain selection

Importance of spiral selector for single crystal casting (effect of take-off angle).



Cylindrical selector (for comparison) does not successfully select one grain after a similar solidification length as the spiral selector.

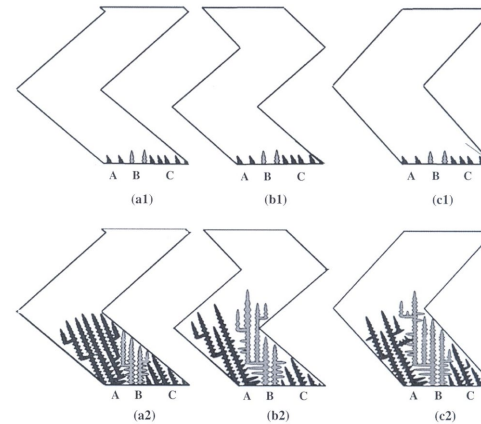
Dan et al., Metal. & Mat. Trans. A, Vol 42A, pp. 3430-3438, 2011

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Grain selection

Schematic diagrams of the geometrical blocking mechanisms (a) at the beginning of solidification, (b) at about a pitch height.



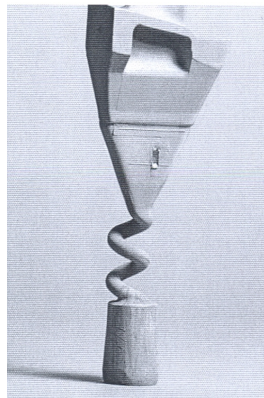
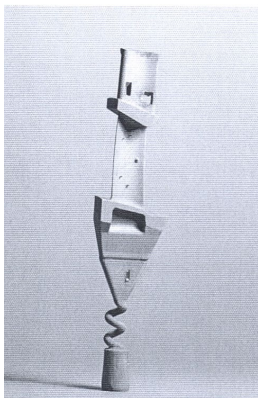
Dan et al., Metal. & Mat. Trans. A, Vol 42A, pp. 3430-3438, 2011

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Example of single crystal blades

- Trent 800 high-pressure blade



Left: after removal of the investment shell.

Right: Details of the starter block and pig-tail grain selector (directional solidification at the starter block).

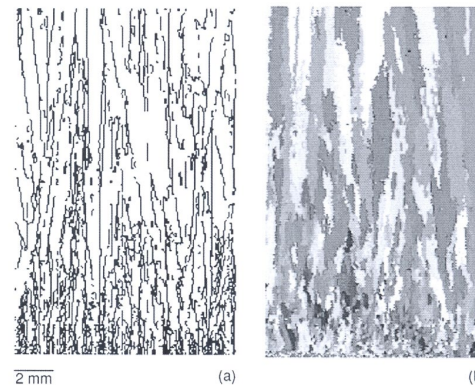
R. Reed, The superalloys: fundamentals and applications, Chapter 3: Single-crystal superalloys for blades applications, Cambridge University Press, 2006

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Example of single crystal blades

- Grain structure in the starter block



Left: theoretical calculations.

Right: Experimental analysis (EBSD).

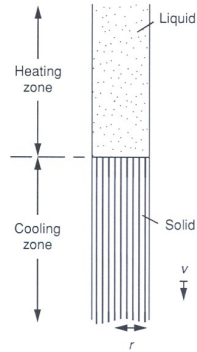
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Analysis of heat transfer during solidification

- Estimation of withdrawal velocity



$$v = \frac{1}{c} \left(\frac{2h_{eff} \kappa_s}{r} \right)^{\frac{1}{2}} (Z^{\frac{1}{2}} - Z^{-\frac{1}{2}})$$

$$Z = \frac{L + c(T_1 - T_0)}{L + c(T_1 - T_m)}$$

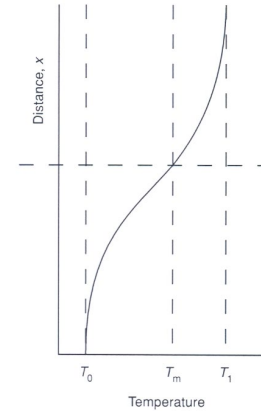
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Analysis of heat transfer during solidification

- Estimation of temperature fields



$$T(x) = T_0 + (T_m - T_0) \exp\{\beta_s x\} \quad x < 0$$

$$T(x) = T_1 - (T_1 - T_m) \exp\{-\beta_l x\} \quad x > 0$$

$$\beta_s = \left[\frac{2h_{eff}}{\kappa_s r} + \left(\frac{vc}{2\kappa_s} \right)^2 \right]^{\frac{1}{2}} - \frac{vc}{\kappa_s} \quad \beta_l = \frac{vc}{\kappa_l}$$

R. Reed, The superalloys: fundamentals and applications, Chapter 3: Single-crystal superalloys for blades applications, Cambridge University Press, 2006

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Analysis of heat transfer during solidification

- Comparison of axial and transverse heat transfer

$$q = h_{eff}(T - T_0) = (h_c + h_R)(T - T_0)$$

$$h_c = \frac{\kappa_s}{X}$$

$$h_R = \frac{\sigma(\varepsilon T^4 - \alpha T_0^4)}{T - T_0}$$

h_c and h_R corresponding to heat transfer coefficients for axial conduction and transverse radiation, respectively.

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Analysis of heat transfer during solidification

- List of parameters:

x = distance in axial direction

c = volumetric specific heat

κ = thermal conductivity (s and l subscripts refer to solid and liquid)

T_m = melting temperature; T_0 = ambient temperature; T_1 = furnace temperature

r = radius of the casting

h_{eff} = heat-transfer coefficient

L = volumetric latent heat of fusion

X = length of solidified ingot

σ = Stefan constant

ε = emissivity of the ingot

α = absorptivity of the surrounding

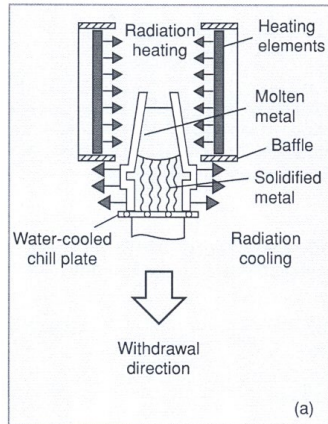
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Effect of quenching

- Conventional Bridgman process (high-rate solidification or HRS)



$$h_{eff}^{Bridgman} = \left[\frac{1}{h_C} + \frac{1}{h_{gap}} + \frac{1}{h_{shell}} + \frac{1}{h_R} \right]^{-1}$$

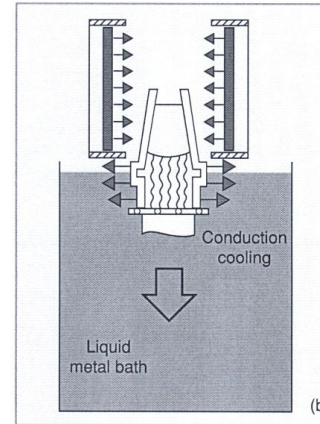
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Effect of quenching

- Liquid metal cooling process



$$h_{eff}^{LM} = \left[\frac{1}{h_C} + \frac{1}{h_{gap}} + \frac{1}{h_{shell}} \right]^{-1}$$

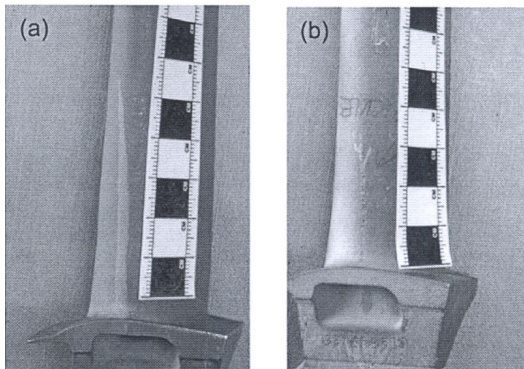
$h_{LMC} \gg h_R$ does not contribute to h_{eff}^{LM}

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Defects in single crystal blades



Left: high angle grain boundaries.

Right: freckle chains.

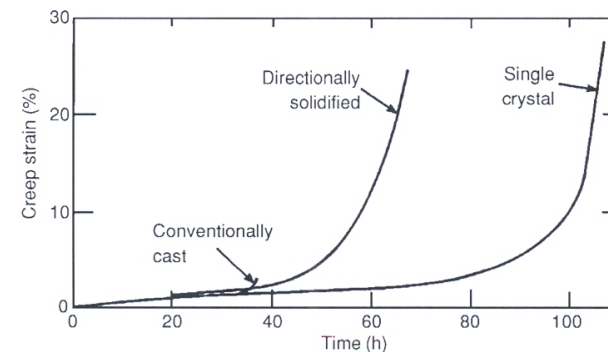
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Creep resistance

- Comparison for superalloy Mar-M200 under loading of 206MPa at 982°C



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